

Applied Chemistry Notebook

Course Code: 1004 | Semester 1 | Credits: 3

Enhanced bilingual study notes with simple explanations, diagrams, solved numerical problems, examples, expected questions and answers.

ലളിതമായ English + Malayalam വിശദീകരണങ്ങൾ, diagrams, solved problems, examples, expected questions & answers എന്നിവ ഉൾപ്പെടുത്തിയ notebook.

Use for:

First revision, exam preparation and quick topic understanding.

Based on:

Applied Chemistry syllabus - atomic structure, analytical chemistry, water treatment, engineering materials, electrochemistry and corrosion.

Learning style:

Read concept - see diagram - study example
- solve problem - revise questions.

Footer:

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Note: This is a student-friendly learning notebook prepared from the provided syllabus. Expected questions are practice-focused, not a guarantee of actual exam questions.

Course Roadmap

What this notebook covers and why it is useful.

English

The syllabus contains four major course outcomes. This notebook follows the same order and adds extra examples, solved problems and model answers for each topic.

Malayalam

സിലബസിൽ നാല് പ്രധാന course outcomes ഉണ്ട്. അതേ order-ൽ ഈ notebook തയ്യാറാക്കിയിരിക്കുന്നു. ഓരോ topic-നും examples, solved problems, model answers എന്നിവ ചേർത്തിട്ടുണ്ട്.

CO	Main area	Study focus	Hours
CO1	Atomic structure and chemical bonding	Models, orbitals, electronic configuration, bonds	8
CO2	Analytical chemistry and water treatment	Molarity, normality, pH, titration, hardness, purification	14
CO3	Engineering materials	Alloys, glasses, refractories, polymers, rubber, nano materials	9
CO4	Electrochemistry and corrosion	Redox, electrolysis, Faraday laws, cells, corrosion control	12

Best method: For every topic, prepare: definition, diagram, example, application, and one solved numerical if applicable.

How to Study This Subject

A simple plan for easy understanding.

English

1. Read the English explanation first.
2. Read the Malayalam helper note to connect the idea.
3. Redraw the diagram once in your notebook.
4. Practice solved problems without looking at the answer.
5. Revise the expected questions before the test.

Malayalam

1. ആദ്യം English explanation വായിക്കുക.
2. Malayalam helper note വായിച്ച് idea connect ചെയ്യുക.
3. Diagram ഒരു തവണ notebook-ൽ വരയ്ക്കുക.
4. Answer നോക്കാതെ solved problems ചെയ്തു നോക്കുക.
5. Test മുൻ expected questions revise ചെയ്യുക.

For 2 marks

Learn definitions + examples.

For 5 marks

Add diagrams and short explanation.

For problems

Write formula, substitute, calculate, final unit.

Presentation tip: In chemistry answers, a neat diagram and one correct example can improve clarity.

Mini Formula Bank

Important formulas used in the numerical problems.

$$\lambda = h / mv$$
 de Broglie wavelength

$$\Delta x \times \Delta p \geq h / 4\pi$$
 uncertainty principle

$$M = \text{moles} / \text{volume in litre}$$
 molarity

$$N = \text{gram equivalents} / \text{litre}$$
 normality

$$\text{ppm} = \text{mg/L}$$
 dilute aqueous solution

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pH} + \text{pOH} = 14$$
 at 25°C

$$N_1V_1 = N_2V_2$$
 normality equation

$$m = EIt / F$$
 Faraday law calculation

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$
 cell emf

Problem rule: Always convert mL to L when using molarity/normality formulas. For electrolysis, convert minutes to seconds.

1

Atomic Structure and Chemical Bonding

Understand atoms, Bohr model, dual nature, orbitals, electronic configuration and bond formation.

Atoms, Bohr model, dual nature, orbitals, electronic configuration, bonding
എന്നിവ മനസ്സിലാക്കുന്ന module.

[Simple notes](#)

[Examples](#)

[Solved problems](#)

[Expected questions](#)

Atom Structure

Basic parts of an atom.

English

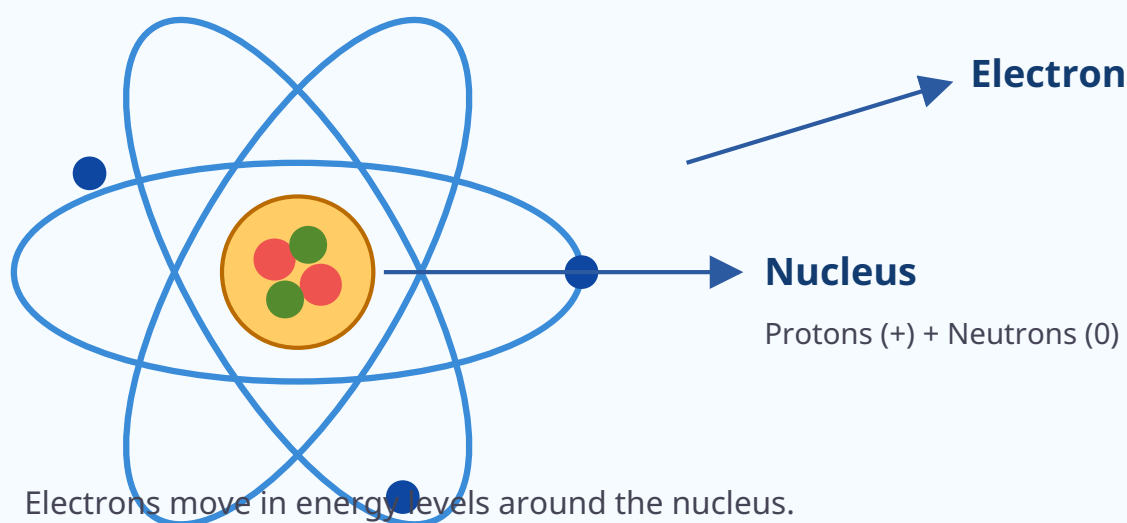
An atom is the smallest unit of an element that retains the chemical property of that element. It contains a nucleus at the centre and electrons outside.

- Proton: positive charge; present in nucleus.
- Neutron: no charge; present in nucleus.
- Electron: negative charge; moves around nucleus.

Malayalam

ഒരു മൂലകത്തിന്റെ രാസ സ്വഭാവം നിലനിർത്തുന്ന ഏറ്റവും ചെറിയ ഘടകമാണ് atom. നടുവിൽ nucleus ഉണ്ട്; പുറത്തായി electrons കാണുന്നു.

- Proton: positive charge; nucleus-ൽ കാണും.
- Neutron: charge ഇല്ല; nucleus-ൽ കാണും.
- Electron: negative charge; nucleus-നെ ചുറ്റി സഞ്ചരിക്കും.



Description: The nucleus contains protons and neutrons. Electrons occupy energy levels around the nucleus.

വിവരണം: Nucleus-ൽ protons, neutrons ഉണ്ട്. Electrons nucleus-നെ ചുറ്റിയുള്ള energy levels-ൽ കാണുന്നു.

Atomic number

Number of protons.

Mass number

Protons + neutrons.

Neutral atom

Protons = electrons.

Bohr Atom Model

A simple model to understand shells and energy levels.

English

Postulates

- Electrons revolve around the nucleus in fixed circular paths called orbits.
- Each orbit has definite energy.
- Electrons do not radiate energy in a permitted orbit.
- Energy is absorbed or emitted when electrons jump between orbits.

Malayalam

പ്രധാന ആശയങ്ങൾ

- Electrons nucleus-നെ ചുറ്റി fixed circular paths-ൽ സഞ്ചരിക്കുന്നു. ഇവ orbits എന്നാണ്.
- ഓരോ orbit-നും definite energy ഉണ്ട്.
- Allowed orbit-ൽ electron energy നഷ്ടപ്പെടുത്തില്ല.
- Orbit മാറുമ്പോൾ energy absorb/emit ചെയ്യും.

Merit

Hydrogen spectrum explained.

Demerit

Cannot explain multi-electron atoms completely.

Exam use

Draw shells and write postulates.

Expected question: State Bohr's postulates. Write any two merits and two demerits.

Dual Nature of Matter

Particles can show wave nature.

English

de Broglie proposed that a moving particle has wave character. The wavelength is called de Broglie wavelength.

$$\lambda = h / mv$$

When mass or velocity increases, wavelength decreases.

Malayalam

de Broglie പ്രകാരം സഞ്ചരിക്കുന്ന particle-ന് wave nature ഉണ്ട്. അതിന്റെ wavelength ആണ് de Broglie wavelength.

$$\lambda = h / mv$$

Mass അല്ലെങ്കിൽ velocity കൂടുമ്പോൾ wavelength കുറയും.

Solved problem 1: Calculate the de Broglie wavelength of an electron moving with velocity 2.0×10^6 m/s.

Given mass of electron = 9.11×10^{-31} kg, $h = 6.626 \times 10^{-34}$ Js.

Formula: $\lambda = h / mv$

Substitute: $\lambda = 6.626 \times 10^{-34} / (9.11 \times 10^{-31} \times 2.0 \times 10^6)$

Answer: $\lambda = 3.64 \times 10^{-10}$ m.

Malayalam help: formula എഴുതുക, values substitute ചെയ്യുക, final unit meter ആയി എഴുതുക.

Try yourself:

A particle has mass 1×10^{-27} kg and velocity 1×10^5 m/s. Find λ . **Answer: 6.626×10^{-12} m.**

Heisenberg Uncertainty Principle

Exact position and momentum cannot both be known at the same time.

English

For microscopic particles like electrons, we cannot determine exact position and exact momentum simultaneously.

$$\Delta x \times \Delta p \geq h / 4\pi$$

Malayalam

Electron പോലുള്ള ചെറിയ particles-ന്റെ exact position-യും exact momentum-വും ഒരേ സമയം കൃത്യമായി അറിയാൻ സാധിക്കില്ല.

$$\Delta x \times \Delta p \geq h / 4\pi$$

Solved problem 2: If uncertainty in position of an electron is 1×10^{-10} m, calculate minimum uncertainty in velocity.

Given: $\Delta x = 1 \times 10^{-10}$ m, $h = 6.626 \times 10^{-34}$ Js, $m = 9.11 \times 10^{-31}$ kg.

$\Delta p = h / 4\pi\Delta x = 5.27 \times 10^{-25}$ kg m/s.

$\Delta v = \Delta p / m = 5.27 \times 10^{-25} / 9.11 \times 10^{-31}$.

Answer: $\Delta v = 5.79 \times 10^5$ m/s.

Malayalam help: ആദ്യം momentum uncertainty കണ്ടെത്തുക; പിന്നെ $\Delta v = \Delta p / m$ ഉപയോഗിക്കുക.

Concept: Smaller uncertainty in position means larger uncertainty in momentum.

Orbitals and Quantum Numbers

The address system of electrons.

English

An orbital is a region around the nucleus where the probability of finding an electron is maximum. Quantum numbers describe the position and state of an electron.

Malayalam

Nucleus-നെ ചുറ്റിയുള്ള electron കാണാനുള്ള probability കൂടുതലുള്ള പ്രദേശമാണ് orbital. Quantum numbers electron-ന്റെ state/position വിവരണം നൽകുന്നു.

Quantum number	Symbol	Meaning	Example
Principal	n	Shell and energy	n = 1, 2, 3
Azimuthal	l	Subshell and shape	s, p, d, f
Magnetic	m	Orientation of orbital	-l to +l
Spin	s	Electron spin direction	+1/2 or -1/2

s orbital
spherical

p orbital
dumbbell

d orbital
complex shape

Expected question: Define orbital. Name four quantum numbers and write their significance.

Electronic Configuration Rules

How electrons are filled in orbitals.

English

- **Aufbau rule:** Lower energy orbitals are filled first.
- **Pauli exclusion principle:** One orbital can contain a maximum of two electrons with opposite spins.
- **Hund rule:** In equal energy orbitals, electrons occupy singly before pairing.

Malayalam

- **Aufbau rule:** Lower energy orbitals ആദ്യം നിറയും.
- **Pauli exclusion principle:** ഒരു orbital-ൽ maximum രണ്ട് electrons; spin opposite ആയിരിക്കും.
- **Hund rule:** Equal energy orbitals-ൽ pairing ചെയ്യുന്നതിന് മുമ്പ് ഓരോന്നായി നിറയും.

Element	Z	Electronic configuration	Shell notation
H	1	1s ¹	1
He	2	1s ²	2
C	6	1s ² 2s ² 2p ²	2,4
O	8	1s ² 2s ² 2p ⁴	2,6
Ne	10	1s ² 2s ² 2p ⁶	2,8
Na	11	1s ² 2s ² 2p ⁶ 3s ¹	2,8,1
Mg	12	1s ² 2s ² 2p ⁶ 3s ²	2,8,2
Cl	17	1s ² 2s ² 2p ⁶ 3s ² 3p ⁵	2,8,7
Ca	20	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ²	2,8,8,2

Electronic Configuration - Examples

Step-by-step writing practice.

Example 1: Write electronic configuration of sodium (Na), atomic number 11.

Number of electrons = 11.

Fill orbitals in order: 1s, 2s, 2p, 3s.

Configuration: 1s² 2s² 2p⁶ 3s¹. Shell notation: 2, 8, 1.

Malayalam help: atomic number = electrons. ആദ്യം 1s², പിന്നെ 2s², 2p⁶, ബാക്കി 3s¹.

Example 2: Write electronic configuration of chlorine (Cl), atomic number 17.

Electrons = 17.

1s² 2s² 2p⁶ uses 10 electrons. Remaining 7 go to 3s² 3p⁵.

Configuration: 1s² 2s² 2p⁶ 3s² 3p⁵. Shell notation: 2, 8, 7.

Practice:

1. Write configuration of Al (Z = 13). **Answer: 1s² 2s² 2p⁶ 3s² 3p¹.**
2. Write configuration of K (Z = 19). **Answer: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s¹.**

Chemical Bonding

Why atoms combine with other atoms.

English

Atoms form chemical bonds to attain stability. The octet rule says that atoms tend to complete eight electrons in their outermost shell.

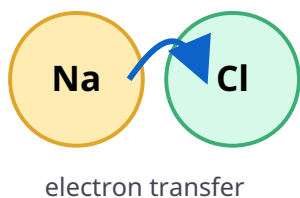
- Ionic bond: electron transfer.
- Covalent bond: electron sharing.
- Coordinate bond: shared pair donated by one atom.

Malayalam

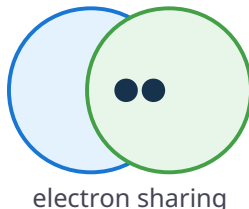
Atoms stability നേടാൻ chemical bond ഉണ്ടാക്കുന്നു. Octet rule പ്രകാരം outermost shell-ൽ 8 electrons നേടാൻ atoms ശ്രമിക്കുന്നു.

- Ionic bond: electron transfer.
- Covalent bond: electron sharing.
- Coordinate bond: shared pair ഒരു atom നൽകുന്നു.

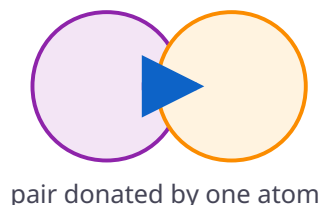
Ionic bond



Covalent bond



Coordinate bond



Description: Ionic bond involves transfer of electrons; covalent bond involves sharing; coordinate bond has a shared pair donated by one atom.

വിവരണം: Ionic bond-ൽ electron transfer, covalent bond-ൽ sharing, coordinate bond-ൽ shared pair ഒരു atom നൽകുന്നതാണ്.

NaCl
ionic

H₂ and HF
covalent

NH₄⁺
coordinate

Bonding Examples and Hydrogen Bonding

Common examples for exam answers.

English

Example: NaCl

Sodium loses one electron to become Na^+ . Chlorine gains one electron to become Cl^- . The electrostatic attraction between Na^+ and Cl^- forms ionic bond.

Example: H_2

Two hydrogen atoms share one pair of electrons to become stable.

Malayalam

Example: NaCl

Sodium ഒരു electron നഷ്ടപ്പെടുത്തി Na^+ ആകുന്നു. Chlorine ഒരു electron സ്വീകരിച്ച് Cl^- ആകുന്നു. Na^+ - Cl^- attraction ആണ് ionic bond.

Example: H_2

രണ്ട് hydrogen atoms ഒരു electron pair share ചെയ്ത് stable ആകുന്നു.

Hydrogen bonding: A weak attraction between hydrogen attached to a highly electronegative atom and another electronegative atom. In water, hydrogen bonding gives high boiling point and unusual behaviour of ice.

Answer line: The anomalous behaviour of H_2O is due to intermolecular hydrogen bonding.

Module 1 - Expected Questions with Answers

Practice questions based on atomic structure and bonding.

1. Define atomic number and mass number.

Atomic number is the number of protons in the nucleus. Mass number is the sum of protons and neutrons.

2. State de Broglie equation.

The wavelength associated with a moving particle is $\lambda = h/mv$, where h is Planck's constant, m is mass and v is velocity.

3. What is Heisenberg uncertainty principle?

It is impossible to know both exact position and exact momentum of a microscopic particle simultaneously. $\Delta x \times \Delta p \geq h/4\pi$.

4. Write any two rules for electronic configuration.

Aufbau rule: electrons fill lower energy orbitals first. Pauli principle: maximum two electrons occupy an orbital with opposite spins.

5. Explain ionic bond with example.

Ionic bond is formed by transfer of electrons. In NaCl, Na gives one electron to Cl and forms Na^+ and Cl^- ions which attract each other.

6. Why does water show anomalous behaviour?

Water shows anomalous behaviour because of hydrogen bonding between water molecules.

Module 1 - More Answer Writing Practice

Use these for 5-mark or 8-mark answers.

Long answer 1: Explain Bohr atom model with merits and demerits.

Write postulates: fixed orbits, definite energy, no energy loss in permitted orbit, energy emitted/absorbed during jumps. Merit: explains hydrogen spectrum. Demerits: cannot explain multi-electron atoms and fine structure completely.

Long answer 2: Compare ionic, covalent and coordinate bonds.

Bond	Formation	Example
Ionic	Transfer of electrons	NaCl
Covalent	Sharing of electrons	H ₂ , HF
Coordinate	Shared pair donated by one atom	NH ₄ ⁺

Numerical practice set

- Find wavelength when $m = 2 \times 10^{-26}$ kg and $v = 3 \times 10^3$ m/s. **1.10×10^{-11} m**
- If $\Delta x = 2 \times 10^{-10}$ m, find Δp . **2.64×10^{-25} kg m/s**

2

Analytical Chemistry and Water Treatment

Learn solution concentration, pH, titration, hardness of water, softening and municipal purification.

Solution concentration, pH, titration, hardness, softening, potable water treatment എന്നിവ പഠിക്കുന്ന module.

[Simple notes](#)

[Examples](#)

[Solved problems](#)

[Expected questions](#)

Solutions and Concentration

Solute, solvent, molarity, normality and ppm.

English

A solution is a homogeneous mixture of solute and solvent. Concentration tells how much solute is present in a given amount of solution.

- Solute: substance dissolved.
- Solvent: substance that dissolves solute.
- Solution: solute + solvent.

Malayalam

Solute, solvent എന്നിവ ചേർന്ന homogeneous mixture ആണ് solution. Concentration എന്നത് solution-ൽ എത്ര solute ഉണ്ട് എന്ന് പറയുന്നു.

- Solute: dissolve ചെയ്യുന്ന വസ്തു.
- Solvent: solute dissolve ചെയ്യുന്ന medium.
- Solution: solute + solvent.

Term	Formula	Unit/meaning
Molarity	$M = \text{moles} / \text{litre}$	mol L ⁻¹
Normality	$N = \text{gram equivalents} / \text{litre}$	eq L ⁻¹
ppm	ppm = mg/L	parts per million for dilute water solution

Molarity - Solved Problems

Practice concentration calculation.

Solved problem 1: 5.85 g NaCl is dissolved in water and volume is made to 500 mL. Find molarity. Molecular mass of NaCl = 58.5 g/mol.

| Moles of NaCl = given mass / molar mass = $5.85 / 58.5 = 0.1$ mol.

| Volume = 500 mL = 0.5 L.

| Molarity = moles / litre = $0.1 / 0.5$.

| **Answer: M = 0.2 M.**

Malayalam help: mL നെ litre ആക്കാൻ 1000-ൽ divide ചെയ്യുക.

Solved problem 2: Find mass of NaOH needed to prepare 250 mL of 0.1 M solution. Molar mass NaOH = 40 g/mol.

| Moles = $M \times V = 0.1 \times 0.25 = 0.025$ mol.

| Mass = moles x molar mass = 0.025×40 .

| **Answer: 1.0 g NaOH.**

Try yourself:

2 g NaOH in 500 mL solution. Find molarity. **Answer: 0.1 M.**

Normality and ppm - Solved Problems

Simple examples for quantitative analysis.

Solved problem 1: Find normality of 4.9 g H₂SO₄ dissolved in 1 litre. Molecular mass = 98, basicity = 2.

| Equivalent weight of H₂SO₄ = molecular mass / basicity = $98 / 2 = 49$.

| Gram equivalents = given mass / equivalent weight = $4.9 / 49 = 0.1$.

| Normality = gram equivalents / litre = $0.1 / 1$.

| **Answer: 0.1 N.**

Solved problem 2: 10 mg of impurity is present in 2 L water. Calculate ppm.

| ppm = mg/L.

| ppm = $10 \text{ mg} / 2 \text{ L}$.

| **Answer: 5 ppm.**

Malayalam help: water solution-ൽ ppm സാധാരണ mg per litre ആയി എഴുതാം.

Memory: Molarity uses molecular mass. Normality uses equivalent weight.

pH and pOH Scale

Acidity and basicity made simple.

English

pH tells whether a solution is acidic, neutral or basic. Acidic solution has pH less than 7. Neutral solution has pH 7. Basic solution has pH greater than 7.

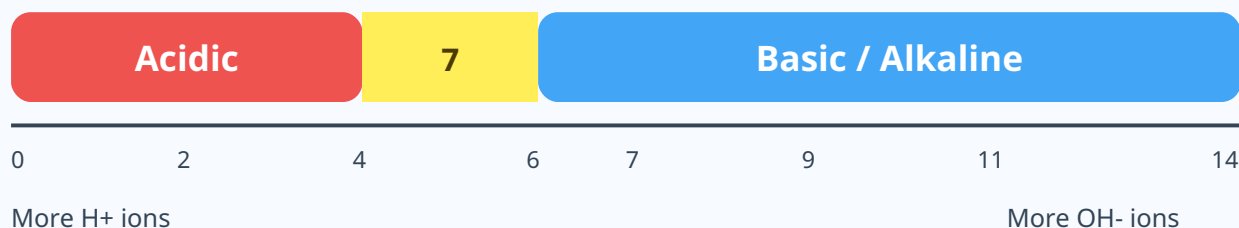
$$\text{pH} = -\log[\text{H}^+] \quad \text{and} \quad \text{pH} + \text{pOH} = 14$$

Malayalam

pH ഒരു solution acidic ആണോ neutral ആണോ basic ആണോ എന്ന് കാണിക്കുന്നു. pH 7-ൽ താഴെ acid, pH 7 neutral, pH 7-നു മുകളിൽ base.

$$\text{pH} = -\log[\text{H}^+] \quad \text{കൂടാതെ} \quad \text{pH} + \text{pOH} = 14$$

pH Scale



Description: pH less than 7 is acidic, pH 7 is neutral and pH greater than 7 is basic.

വിവരണം: pH 7-ൽ താഴെ acidic, pH 7 neutral, pH 7-നു മുകളിൽ basic ആണ്.

pH and pOH - Solved Problems

Step-by-step logarithm problems.

Solved problem 1: Calculate pH of a solution having $[H^+] = 1 \times 10^{-3}$ M.

| Formula: $pH = -\log[H^+]$

| $pH = -\log(1 \times 10^{-3})$

| **Answer: pH = 3. The solution is acidic.**

Solved problem 2: If pOH = 5, find pH.

| Formula: $pH + pOH = 14$.

| $pH = 14 - 5$.

| **Answer: pH = 9. The solution is basic.**

Solved problem 3: Calculate pOH if $[OH^-] = 1 \times 10^{-4}$ M.

| $pOH = -\log[OH^-] = -\log(10^{-4})$

| **pOH = 4, pH = 10.**

Buffer Solution

Solution that resists change in pH.

English

A buffer solution resists change in pH when a small amount of acid or base is added.

- Acidic buffer: weak acid + salt of weak acid with strong base. Example: acetic acid + sodium acetate.
- Basic buffer: weak base + salt of weak base with strong acid. Example: ammonium hydroxide + ammonium chloride.

Malayalam

ചെറിയ അളവിൽ acid/base ചേർത്താലും pH വലിയ രീതിയിൽ മാറാതെ നിലനിർത്തുന്ന solution ആണ് buffer.

- Acidic buffer: weak acid + അതിന്റെ salt. Example: acetic acid + sodium acetate.
- Basic buffer: weak base + അതിന്റെ salt. Example: ammonium hydroxide + ammonium chloride.

Use

Maintains pH in labs.

Biological use

Blood pH control.

Exam key

Write definition + one example each.

Volumetric Analysis and Titration

Finding concentration using reaction volume.

English

Titration is a method of finding unknown concentration by reacting it with a standard solution. The end point is the stage where the indicator changes colour.

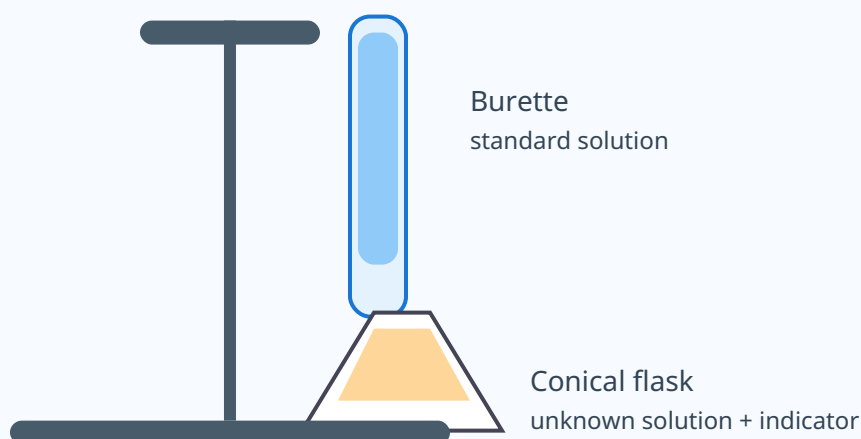
$$N_1V_1 = N_2V_2$$

Malayalam

Known concentration ഉള്ള standard solution ഉപയോഗിച്ച് unknown concentration കണ്ടെത്തുന്ന method ആണ് titration. Indicator colour മാറുന്ന stage ആണ് end point.

$$N_1V_1 = N_2V_2$$

Titration Setup



Description: Standard solution is taken in a burette and added drop by drop to the unknown solution in the conical flask.

വിവരണം: Burette-ൽ standard solution എടുത്ത് conical flask-ിലുള്ള unknown solution-ലേക്ക് drop by drop ചേർക്കുന്നു.

Indicator choice: Choose an indicator whose colour change occurs near the pH at the equivalence point.

Titration - Solved Problem

Normality equation practice.

Solved problem: 20 mL of acid is completely neutralized by 25 mL of 0.1 N NaOH. Find normality of the acid.

Use normality equation: $N_1V_1 = N_2V_2$.

Let acid normality = N_1 , acid volume $V_1 = 20$ mL.

Base normality $N_2 = 0.1$ N, base volume $V_2 = 25$ mL.

$N_1 \times 20 = 0.1 \times 25$.

$N_1 = 2.5 / 20 = 0.125$.

Answer: Acid normality = 0.125 N.

Malayalam help: Acid-ന്റെ normality unknown ആകെ $N_1V_1 = N_2V_2$ substitute ചെയ്യുക.

Try yourself:

25 mL HCl is neutralized by 20 mL of 0.2 N NaOH. Find normality of HCl. **Answer: 0.16 N.**

Hardness of Water

Why soap does not lather easily in hard water.

English

Hard water does not give lather easily with soap because it contains calcium and magnesium salts. These ions form insoluble scum with soap.

- Temporary hardness: due to bicarbonates of Ca and Mg; removed by boiling.
- Permanent hardness: due to chlorides/ sulphates of Ca and Mg; not removed by boiling.

Malayalam

Calcium, magnesium salts ഉള്ളതിനാൽ hard water soap-നൊപ്പം എളുപ്പത്തിൽ lather ഉണ്ടാക്കില്ല. ഇവ soap-നൊപ്പം insoluble scum ഉണ്ടാക്കുന്നു.

- Temporary hardness: Ca/Mg bicarbonates മൂലം; boiling വഴി കുറയ്ക്കാം.
- Permanent hardness: Ca/Mg chlorides/ sulphates മൂലം; boiling കൊണ്ട് പോകില്ല.

Boiler problems: Hard water causes scale formation, reduced heat transfer, wastage of fuel and corrosion.

Type	Cause	Removal
Temporary hardness	$\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$	Boiling or lime treatment
Permanent hardness	CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4	Soda-lime process or ion exchange

Water Softening Methods

Soda-lime process and ion exchange method.

English

Soda-lime process

Lime and soda are added to hard water. Calcium and magnesium ions are converted into insoluble precipitates and removed.

Ion exchange method

Hardness-causing ions Ca^{2+} and Mg^{2+} are exchanged with Na^{+} or H^{+} ions using ion exchange resins.

Malayalam

Soda-lime process

Hard water-ലേക്ക് lime, soda ചേർത്ത് Ca^{2+} , Mg^{2+} ions insoluble precipitate ആക്കി remove ചെയ്യുന്നു.

Ion exchange method

Ion exchange resins ഉപയോഗിച്ച് Ca^{2+} , Mg^{2+} ions, Na^{+} അല്ലെങ്കിൽ H^{+} ions-നോട് exchange ചെയ്യുന്നു.

Soft water

Gives lather easily.

Ion exchange

High quality water.

Boilers

Soft water prevents scale.

Expected question: Explain ion exchange method for softening hard water.

Municipal Water Treatment

How potable water is supplied.

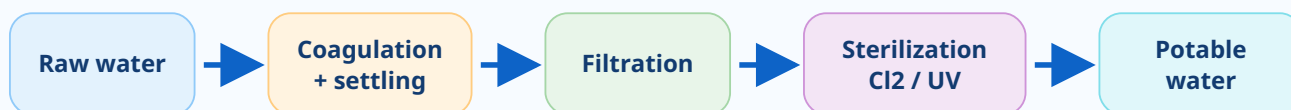
English

Potable water is water fit for drinking. It should be clear, colourless, odourless, free from harmful microorganisms and safe for domestic use.

Malayalam

കുടിക്കാൻ സുരക്ഷിതമായ വെള്ളമാണ് potable water. അത് clear, colourless, odourless, harmful microorganisms ഇല്ലാത്തത് ആയിരിക്കണം.

Municipal Water Treatment Flow



Goal: remove particles, reduce colour/odour and kill disease-causing germs.

Description: Municipal treatment usually includes sedimentation, coagulation, filtration and sterilization.

വിവരണം: Municipal water treatment-ൽ sedimentation, coagulation, filtration, sterilization തുടങ്ങിയ steps ഉണ്ട്.

Step	Purpose
Sedimentation	Heavy suspended particles settle down.
Coagulation	Fine particles combine into larger flocs.
Filtration	Removes remaining suspended impurities.
Sterilization	Kills germs using chlorination/bleaching/UV radiation.

Module 2 - Expected Questions with Answers

Practice questions based on analytical chemistry.

1. Define molarity and normality.

Molarity is the number of moles of solute present in one litre of solution. Normality is the number of gram equivalents of solute present in one litre of solution.

2. What is pH? What is the pH of neutral water?

pH is the negative logarithm of hydrogen ion concentration. $\text{pH} = -\log[\text{H}^+]$. Neutral water has pH 7 at 25°C.

3. Define buffer solution. Give one example.

A buffer solution resists change in pH when small amount of acid or base is added. Example: acetic acid + sodium acetate.

4. What is end point in titration?

End point is the stage in titration where the indicator changes colour, showing completion of reaction.

5. Why does hard water not lather with soap?

Ca^{2+} and Mg^{2+} ions in hard water react with soap to form insoluble scum, so lather formation is poor.

Module 2 - Long Answers and Problems

Answer patterns for exams.

Long answer 1: Explain municipal water treatment.

Write the flow: raw water - sedimentation - coagulation - filtration - sterilization - potable water. Explain each step in one or two sentences and draw the flow chart.

Long answer 2: Compare temporary and permanent hardness.

Temporary hardness	Permanent hardness
Due to bicarbonates of Ca/Mg.	Due to chlorides/sulphates of Ca/Mg.
Removed by boiling.	Not removed by boiling.
Example: $\text{Ca}(\text{HCO}_3)_2$.	Example: CaSO_4 .

Numerical practice set

- Find pH when $[\text{H}^+] = 1 \times 10^{-5} \text{ M}$. **pH = 5**
- Find pH when $\text{pOH} = 3$. **pH = 11**
- 25 mL acid reacts with 25 mL of 0.1 N base. Find acid normality. **0.1 N**

3

Engineering Materials

Study alloys, glasses, refractory materials, polymers, rubber and nanomaterials used in engineering.

Engineering applications-ൽ ഉപയോഗിക്കുന്ന alloys, glasses, refractories, polymers, rubber, nanomaterials എന്നിവ പഠിക്കുന്നു.

[Simple notes](#)

[Examples](#)

[Solved problems](#)

[Expected questions](#)

Alloys

Purpose and applications of alloying.

English

An alloy is a homogeneous mixture of two or more metals, or a metal with a non-metal. Alloying is done to improve properties such as strength, hardness, corrosion resistance and melting point.

Malayalam

രണ്ട് അല്ലെങ്കിൽ കൂടുതൽ metals, അല്ലെങ്കിൽ metal + non-metal ചേർന്ന homogeneous mixture ആണ് alloy. Strength, hardness, corrosion resistance, melting point എന്നിവ മെച്ചപ്പെടുത്താൻ alloying ചെയ്യുന്നു.

Alloy	Composition	Application
Brass	Copper + Zinc	Screws, taps, decorative items, instruments
Bronze	Copper + Tin	Bearings, statues, coins
Solder	Lead + Tin	Joining electrical wires and electronic components

Expected question: Define alloy. Write any three purposes of alloying with examples.

Glasses and Refractory Materials

Materials used in high-temperature and engineering applications.

English

Glass

Glass is an amorphous, hard, brittle and transparent material. Types include sodium silicate glass, borosilicate glass, safety glass and insulating glass.

Refractory material

A refractory is a material that can withstand very high temperature without softening or melting.

Malayalam

Glass

Glass ഒരു amorphous, hard, brittle, transparent material ആണ്. Sodium silicate glass, borosilicate glass, safety glass, insulating glass എന്നിവ പ്രധാന types ആണ്.

Refractory material

വളരെ ഉയർന്ന temperature-ൽ soft/melt ചെയ്യാതെ നിലനിൽക്കുന്ന material ആണ് refractory.

Material	Special property	Use
Borosilicate glass	Thermal shock resistance	Laboratory glassware
Safety glass	Does not break into sharp pieces easily	Vehicle windshields
Fire clay / silica brick	High temperature resistance	Furnace lining

Polymers and Polymerization

Large molecules made from monomers.

English

A polymer is a large molecule formed by joining many small molecules called monomers. Polymerization is the process of formation of polymer from monomers.

- Homopolymer: made from one type of monomer.
- Copolymer: made from two or more different monomers.
- Addition polymer: monomers add without loss of small molecules.
- Condensation polymer: small molecules like water are eliminated.

Malayalam

Monomers എന്ന ചെറിയ molecules പലതും ചേർന്ന് ഉണ്ടാകുന്ന വലിയ molecule ആണ് polymer. Monomers ചേർന്ന് polymer ഉണ്ടാകുന്ന process ആണ് polymerization.

- Homopolymer: ഒരേ monomer type.
- Copolymer: രണ്ട് അല്ലെങ്കിൽ കൂടുതൽ monomer types.
- Addition polymer: small molecule eliminate ചെയ്യാതെ addition.
- Condensation polymer: water പോലുള്ള small molecules പുറത്തു പോകും.

Polymerization



Small monomers



Long chain polymer

Description: Small monomers join together to form a long-chain polymer.

വിവരണം: ചെറിയ monomers ചേർന്ന് long-chain polymer രൂപപ്പെടുന്നു.

Common Polymers

Monomers and uses for exam revision.

Polymer	Monomer(s)	Type/use
Polythene	Ethene	Plastic bags, containers, insulation
PVC	Vinyl chloride	Pipes, cable insulation, floor coverings
Nylon-66	Hexamethylene diamine + adipic acid	Fibres, ropes, gears
Bakelite	Phenol + formaldehyde	Switches, handles, electrical fittings

English

Thermoplastics soften on heating and harden on cooling repeatedly. Example: polythene, PVC. Thermosetting plastics become permanently hard on heating. Example: Bakelite.

Malayalam

Thermoplastics heat ചെയ്യുമ്പോൾ soften ആകുകയും cooling-ൽ hard ആകുകയും വീണ്ടും വീണ്ടും ആവർത്തിക്കാം. Example: polythene, PVC. Thermosetting plastics heat ചെയ്താൽ permanent hard ആകും. Example: Bakelite.

Exam memory: PVC = vinyl chloride; Nylon-66 = two monomers; Bakelite = phenol + formaldehyde.

Rubber

Natural rubber, vulcanization and synthetic rubber.

English

Natural rubber is obtained from latex. It is elastic but becomes soft in hot weather and brittle in cold weather. Vulcanization is heating rubber with sulphur to improve strength, elasticity and resistance.

Synthetic rubbers are man-made rubbers.
Examples: Buna-S and Buna-N.

Malayalam

Natural rubber latex-ൽ നിന്ന് ലഭിക്കുന്നു. അത് elastic ആണ്, പക്ഷേ ചൂടിൽ soft ആകുകയും തണുപ്പിൽ brittle ആകുകയും ചെയ്യും. Sulphur ചേർത്ത് heat ചെയ്യുന്നതാണ് vulcanization; ഇത് strength, elasticity, resistance വർദ്ധിപ്പിക്കും.

Synthetic rubber മനുഷ്യൻ നിർമ്മിക്കുന്ന rubber ആണ്. Examples: Buna-S, Buna-N.

Rubber	Monomers	Use
Buna-S	Butadiene + styrene	Tyres, footwear, cable insulation
Buna-N	Butadiene + acrylonitrile	Oil-resistant hoses, gaskets

Nanomaterials and Nanotechnology

Materials at extremely small scale.

English

Nanomaterials are materials having at least one dimension in the range of about 1 to 100 nm. Nanotechnology deals with design and use of materials at nanoscale.

Malayalam

ഏകദേശം 1 മുതൽ 100 nm വരെ dimension ഉള്ള materials ആണ് nanomaterials. Nanoscale-ൽ materials design/use ചെയ്യുന്ന field ആണ് nanotechnology.

Nanomaterials by Dimension



0D: nanoparticles



1D: nanotube



2D: graphene sheet

Description: 0D, 1D and 2D nanomaterials differ based on how many dimensions are outside the nanoscale.

വിവരണം: 0D, 1D, 2D nanomaterials dimension അടിസ്ഥാനത്തിൽ വ്യത്യസ്തമാണ്.

Type	Example	Application
0D	Nanoparticles	Catalysts, medicine, sensors
1D	Carbon nanotubes	Strong materials, electronics
2D	Graphene	Conductive films, composites

Module 3 - Expected Questions with Answers

Practice questions based on engineering materials.

1. Define alloy. Give two examples.

An alloy is a homogeneous mixture of two or more metals or a metal with a non-metal. Examples: brass and bronze.

2. Write the composition and use of solder.

Solder is an alloy of lead and tin. It is used for joining electrical wires and electronic components.

3. What is a refractory material?

A refractory material can withstand very high temperature without melting or softening. It is used for furnace lining.

4. Define polymer and monomer.

A polymer is a large molecule formed by joining many monomers. A monomer is a small molecule that can join with similar molecules to form polymer.

5. What is vulcanization?

Vulcanization is heating natural rubber with sulphur to improve strength, elasticity and resistance to temperature changes.

6. Define nanotechnology.

Nanotechnology is the science and technology of designing, producing and using materials at nanoscale.

Module 3 - Long Answers and Examples

Use tables and examples to score marks.

Long answer 1: Classify polymers with examples.

Classify as homopolymer/copolymer, addition/condensation, thermoplastic/thermosetting. Give examples: polythene as addition homopolymer, Nylon-66 as condensation polymer, PVC as thermoplastic, Bakelite as thermosetting plastic.

Long answer 2: Write short notes on carbon nanotubes, fullerenes and graphene.

Carbon nanotubes are cylindrical carbon nanostructures (SWCNT/MWCNT). Fullerenes are cage-like carbon molecules. Graphene is a single layer of carbon atoms arranged in a hexagonal sheet. They are useful in electronics, composites and sensors.

Application practice

1. Which alloy is used for bearings? **Bronze**
2. Which glass is preferred for laboratory glassware? **Borosilicate glass**
3. Which rubber resists oil? **Buna-N**

4

Electrochemistry and Corrosion

Study redox reactions, electrolysis, Faraday laws, electroplating, Daniell cell and corrosion prevention.

Redox reactions, electrolysis, Faraday laws, electroplating, Daniell cell, corrosion prevention എന്നിവ പഠിക്കുന്നു.

[Simple notes](#)

[Examples](#)

[Solved problems](#)

[Expected questions](#)

Oxidation, Reduction and Redox

Electron transfer made simple.

English

Oxidation is loss of electrons. Reduction is gain of electrons. A redox reaction is a reaction in which oxidation and reduction occur simultaneously.

OIL RIG means Oxidation Is Loss, Reduction Is Gain.

Malayalam

Oxidation എന്നത് electrons നഷ്ടപ്പെടൽ.
Reduction എന്നത് electrons സ്വീകരിക്കൽ.
Oxidation-വും reduction-വും ഒരേ സമയം നടക്കുന്ന reaction ആണ് redox reaction.

OIL RIG ഓർക്കുക: Oxidation Is Loss, Reduction Is Gain.

Example: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$

| Zn loses two electrons: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$. This is oxidation.

| Cu^{2+} gains two electrons: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. This is reduction.

| **Overall reaction is redox.**

Term	Meaning	Example
Metallic conductor	Conducts by electrons	Copper wire
Electrolytic conductor	Conducts by ions	NaCl solution
Insulator	Does not conduct easily	Rubber

Electrolysis and Electrolytic Cell

Chemical change using electric current.

English

Electrolysis is the decomposition of an electrolyte by passing electric current through it. The device used is called electrolytic cell.

- Anode: positive electrode in electrolysis; oxidation occurs.
- Cathode: negative electrode in electrolysis; reduction occurs.
- Electrolyte: substance that conducts electricity in molten or aqueous state.

Malayalam

Electric current pass ചെയ്താൽ electrolyte decompose ചെയ്യുന്ന process ആണ് electrolysis. ഉപയോഗിക്കുന്ന setup ആണ് electrolytic cell.

- Anode: electrolysis-ൽ positive electrode; oxidation നടക്കും.
- Cathode: electrolysis-ൽ negative electrode; reduction നടക്കും.
- Electrolyte: molten/aqueous state-ൽ electricity conduct ചെയ്യുന്ന substance.

Strong electrolyte

Almost fully ionized.

Weak electrolyte

Partly ionized.

Non-electrolyte

Does not produce ions.

Expected question: Define electrolysis, electrolyte, strong electrolyte and weak electrolyte with examples.

Faraday Laws of Electrolysis

Relation between electricity and deposited mass.

English

First law: The mass of substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed.

Second law: When the same quantity of electricity passes through different electrolytes, masses deposited are proportional to their equivalent weights.

$$m = EIt / F$$

Malayalam

First law: Electrode-ൽ deposit/liberate ചെയ്യുന്ന mass, passed electricity-ക്ക് directly proportional ആണ്.

Second law: ഒരേ quantity electricity വ്യത്യസ്ത electrolytes-ൽ pass ചെയ്താൽ deposited masses അവരുടെ equivalent weights-ന് proportional ആണ്.

$$m = EIt / F$$

Symbol	Meaning
m	Mass deposited in gram
E	Equivalent weight
I	Current in ampere
t	Time in seconds
F	Faraday constant = 96500 C/mol

Faraday Law - Solved Problems

Electrolysis numerical practice.

Solved problem 1: Calculate mass of copper deposited when 2 A current is passed for 30 minutes. Equivalent weight of copper = 31.75 g. $F = 96500$ C.

Formula: $m = EIt / F$.

Convert time: 30 min = 30 x 60 = 1800 s.

$m = 31.75 \times 2 \times 1800 / 96500$.

Answer: $m = 1.18$ g copper.

Malayalam help: Time seconds ആക്കാൻ മറക്കരുത്.

Solved problem 2: Find mass of silver deposited by 0.5 A current for 20 minutes. Equivalent weight of Ag = 107.9 g.

$t = 20 \times 60 = 1200$ s.

$m = 107.9 \times 0.5 \times 1200 / 96500$.

Answer: $m = 0.671$ g silver.

Try yourself:

Find mass deposited when $E = 32$ g, $I = 1$ A, $t = 965$ s. **Answer: 0.32 g.**

Industrial Applications of Electrolysis

Electroplating and electrolytic refining.

English

Electroplating is coating one metal over another by electrolysis. It improves appearance and corrosion resistance.

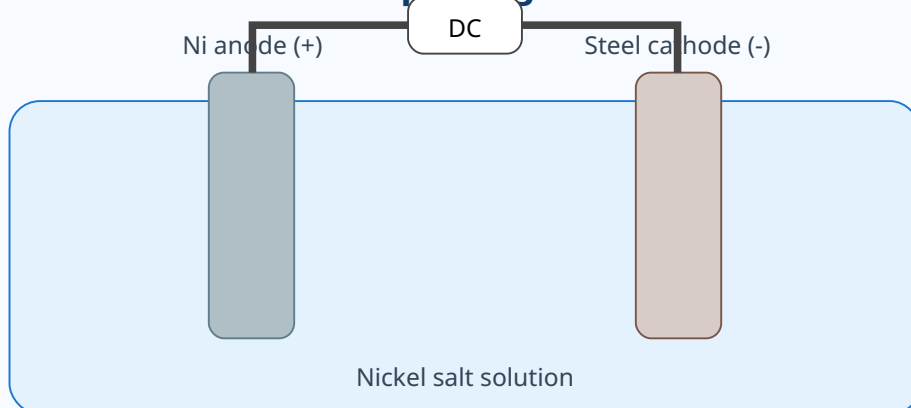
Electrolytic refining is purification of metals using electrolysis. Impure metal acts as anode and pure metal deposits on cathode.

Malayalam

Electroplating electrolysis ഉപയോഗിച്ച് ഒരു metal-ി ന്റെ മേൽ മറ്റൊരു metal coating നൽകുന്ന process ആണ്. Appearance, corrosion resistance മെച്ചപ്പെടുത്തുന്നു.

Electrolytic refining electrolysis ഉപയോഗിച്ച് metal purify ചെയ്യുന്ന process ആണ്. Impure metal anode, pure metal cathode ആയി ഉപയോഗിക്കുന്നു.

Nickel Electroplating on Mild Steel



Metal ions deposit as a thin protective/decorative layer on the cathode.

Description: In nickel plating, the article to be plated is made the cathode and nickel metal is made the anode.

വിവരണം: Nickel plating-ൽ coating ചെയ്യേണ്ട article cathode ആക്കും; nickel metal anode ആക്കും.

Electrochemical Cells and Daniell Cell

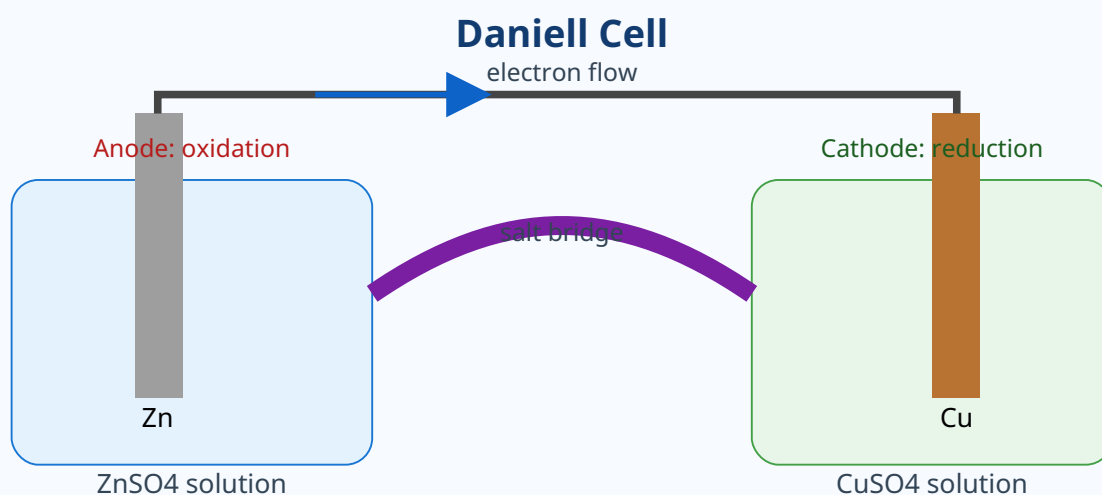
Cells convert chemical energy into electrical energy.

English

An electrochemical cell is a device that converts chemical energy into electrical energy using redox reaction. Cells may be primary cells, secondary cells or fuel cells.

Malayalam

Redox reaction ഉപയോഗിച്ച് chemical energy-യെ electrical energy ആക്കി മാറ്റുന്ന device ആണ് electrochemical cell. Primary, secondary, fuel cells എന്നീ types ഉണ്ട്.



Description: In Daniell cell, zinc acts as anode and copper acts as cathode. Electrons flow through external circuit from zinc to copper.

വിവരണം: Daniell cell-ൽ zinc anode, copper cathode. Electrons external circuit വഴി zinc-ൽ നിന്ന് copper-ലേക്ക് ഒഴുകുന്നു.

Cell type	Feature	Example
Primary cell	Not rechargeable	Dry cell
Secondary cell	Rechargeable	Lead-acid battery
Fuel cell	Continuous supply of fuel	Hydrogen-oxygen fuel cell

Electrochemical Series and EMF

Calculate cell voltage using electrode potentials.

English

The electrochemical series arranges elements according to their standard electrode potentials. The emf of a cell can be calculated from cathode and anode potentials.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

Malayalam

Standard electrode potentials അടിസ്ഥാനത്തിൽ elements arrange ചെയ്തതാണ് electrochemical series. Cathode, anode potentials ഉപയോഗിച്ച് cell emf കണക്കാക്കാം.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

Solved problem: Calculate emf of Daniell cell. $E^{\circ}\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$, $E^{\circ}\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$.

Cathode is Cu; anode is Zn.

$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$.

$E_{\text{cell}} = 0.34 - (-0.76) = 0.34 + 0.76$.

Answer: $E_{\text{cell}} = 1.10 \text{ V}$.

Sign care: Subtracting a negative value means adding it.

Corrosion of Metals

Slow destruction of metals by environment.

English

Corrosion is the gradual destruction of a metal due to chemical or electrochemical reaction with the environment. Rusting of iron is a common example.

Factors affecting corrosion include moisture, oxygen, salts, pH, temperature and nature of metal.

Malayalam

Environment-നൊപ്പം chemical/electrochemical reaction മൂലം metal gradually destroy ചെയ്യുന്നതാണ് corrosion. Iron rusting സാധാരണ example ആണ്.

Moisture, oxygen, salts, pH, temperature, metal nature എന്നിവ corrosion rate-നെ ബാധിക്കുന്നു.

Engineering importance: Corrosion weakens structures, damages machines and increases maintenance cost.

Factor	Effect
Moisture and oxygen	Increase rusting of iron
Salts	Increase conductivity and corrosion rate
Acidic medium	Usually increases corrosion
Protective oxide layer	Can reduce corrosion

Corrosion Prevention

Barrier protection and cathodic protection.

English

Corrosion can be prevented by separating metal from the environment or by making it cathodic.

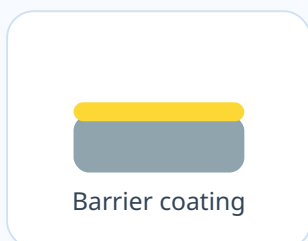
- Barrier protection: painting, oiling, coating.
- Metal coatings: anodic or cathodic coating.
- Non-metallic protection: anodising and anti-rust solution.
- Cathodic protection: sacrificial anode method.

Malayalam

Metal-നെ environment-ൽ നിന്ന് separate ചെയ്യുകയോ metal-നെ cathode ആക്കുകയോ ചെയ്ത് corrosion തടയാം.

- Barrier protection: painting, oiling, coating.
- Metal coatings: anodic/cathodic coating.
- Non-metallic protection: anodising, anti-rust solution.
- Cathodic protection: sacrificial anode method.

Corrosion Prevention



Description: Paint/coating blocks air and moisture. Sacrificial anode corrodes instead of the protected metal.

വിവരണം: Paint/coating air/moisture തടയും. Sacrificial anode protected metal-നു പകരം corrode ചെയ്യും.

Expected question: Explain external corrosion preventive methods with examples.

Module 4 - Expected Questions with Answers

Practice questions based on electrochemistry and corrosion.

1. Define oxidation and reduction using electronic concept.

Oxidation is loss of electrons. Reduction is gain of electrons.

2. What is electrolysis?

Electrolysis is the decomposition of an electrolyte by passing electric current through its molten or aqueous solution.

3. State Faraday's first law of electrolysis.

The mass of substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte.

4. What is electroplating? Give one use.

Electroplating is coating one metal over another using electrolysis. It is used for decorative coating and corrosion protection.

5. Define corrosion.

Corrosion is gradual destruction of metal by chemical or electrochemical reaction with its environment.

6. What is sacrificial anode method?

It is a cathodic protection method in which a more reactive metal is connected to the protected metal, so the reactive metal corrodes instead.

Module 4 - Long Answers and Problems

Answer patterns for exams.

Long answer 1: Explain Daniell cell.

Draw diagram with Zn/ZnSO₄ half-cell and Cu/CuSO₄ half-cell connected by salt bridge. Zinc is anode where oxidation occurs: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$. Copper is cathode where reduction occurs: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. Electrons flow from Zn to Cu through external circuit.

Long answer 2: Explain electroplating of nickel on mild steel.

Clean mild steel article is made cathode. Nickel plate is made anode. Nickel salt solution is used as electrolyte. On passing DC current, Ni²⁺ ions deposit on mild steel as nickel coating.

Numerical practice set

1. Mass of copper deposited when $E = 31.75$, $I = 1 \text{ A}$, $t = 965 \text{ s}$. **0.3175 g**
2. Calculate emf when $E_{\text{cathode}} = +0.80 \text{ V}$ and $E_{\text{anode}} = -0.44 \text{ V}$. **1.24 V**
3. Find time to deposit 0.5 g metal if $E = 25$, $I = 2 \text{ A}$. **$t = 965 \text{ s}$**

Expected Question Bank - Short Answers

Quick revision list for all modules.

Define orbital.

Region where probability of finding electron is maximum.

What is octet rule?

Atoms tend to complete 8 electrons in valence shell.

What is ppm?

Parts per million; for dilute water solution, ppm = mg/L.

What is potable water?

Water fit and safe for drinking.

What is polymerization?

Process of forming polymer from monomers.

What is graphene?

A 2D sheet of carbon atoms arranged in hexagonal pattern.

What is electrolyte?

A substance conducting electricity by ions in molten/ aqueous state.

What is cathodic protection?

Corrosion prevention by making protected metal act as cathode.

Expected Long Questions

Prepare these answer structures.

1. Explain Bohr atom model with postulates, merits and demerits.
2. Explain electronic configuration rules with examples.
3. Compare ionic, covalent and coordinate bonds with examples.
4. Explain molarity, normality and ppm with simple problems.
5. Explain pH scale and solve pH/pOH problems.
6. Describe titration setup and normality equation.
7. Explain causes, types and removal of hardness of water.
8. Draw and explain municipal water treatment flow chart.
9. Classify polymers and explain common polymers with uses.
10. Explain vulcanization and synthetic rubbers.
11. Explain Faraday laws and solve electrolysis problems.
12. Draw and explain Daniell cell.
13. Explain electroplating and electrolytic refining.
14. Explain corrosion and its prevention methods.

Answer writing formula: Definition + principle + diagram/table + example + application + conclusion.

Model Practice Paper with Answer Hints

Use this after completing revision.

Question	Answer hint
1. State de Broglie equation and explain symbols.	$\lambda = h/mv$; define h, m, v.
2. Write electronic configuration of Ca.	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² ; shell 2,8,8,2.
3. Calculate pH when $[H^+] = 10^{-4}$ M.	pH = 4.
4. 10 mL acid reacts with 20 mL 0.1 N base. Find acid normality.	$N_1 \times 10 = 0.1 \times 20$; $N_1 = 0.2$ N.
5. Explain ion exchange method.	Ca ²⁺ /Mg ²⁺ exchange with Na ⁺ /H ⁺ using resin.
6. Write composition and use of brass and solder.	Brass: Cu+Zn; Solder: Pb+Sn.
7. Define nanomaterials and give examples of 0D, 1D, 2D.	1-100 nm; nanoparticles, nanotubes, graphene.
8. State Faraday first law and solve $m = EIt/F$.	Mass proportional to quantity of electricity.
9. Draw Daniell cell and mark anode/cathode.	Zn anode; Cu cathode; salt bridge; electron flow.
10. Explain sacrificial anode method.	More reactive metal corrodes and protects main metal.

Study target: Try to answer all questions without looking at the hints. Then check and improve.

Final Revision Checklist

Tick after completing each task.

- ☐ I can draw atom structure and label proton, neutron and electron.
- ☐ I can state Bohr postulates and write merits/demerits.
- ☐ I can solve de Broglie and uncertainty principle problems.
- ☐ I can write electronic configuration of first 20 elements.
- ☐ I can explain ionic, covalent and coordinate bonds with examples.
- ☐ I can calculate molarity, normality, ppm, pH and pOH.
- ☐ I can explain titration setup and solve $N_1V_1 = N_2V_2$ problems.
- ☐ I can explain hardness of water and softening methods.
- ☐ I can draw municipal water treatment flow chart.
- ☐ I can explain alloys, glass, refractories, polymers and rubber.
- ☐ I can define nanomaterials and write applications.
- ☐ I can state Faraday laws and solve electrolysis problems.
- ☐ I can draw Daniell cell and calculate emf.
- ☐ I can explain corrosion and prevention methods.

Prepared as an enhanced learning notebook for Applied Chemistry 1004.